

BENGALURU, India

WMO: 432950

Lat: 12.967N

Lon: 77.583E

Elev: 921

StdP: 90.74

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.5	16.2	1.6	4.7	28.6	3.9	5.6	27.8	4.4	20.4	4.0	21.3	1.1	90	0.226

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.3	34.3	20.0	33.6	19.9	32.7	19.9	23.7	29.1	23.2	28.6	22.8	28.0	1.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	19.0	25.7	21.8	18.4	25.2	21.4	18.0	24.8	76.3	29.2	74.3	28.8	72.7	28.1	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.3	3.6	13.7	36.1	0.9	1.0	13.0	36.8	12.5	37.3	12.0	37.9	11.4	38.6		
(5)			10.7	25.5	1.0	1.2	10.0	26.4	9.4	27.0	8.8	27.7	8.1	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	22.1	24.2	26.7	27.9	27.0	24.9	23.9	23.7	24.0	23.8	22.6	21.6	(6)
(7)		DBStd	2.26	1.15	1.49	1.53	1.46	1.58	1.28	0.99	0.92	0.99	1.12	1.21	1.19	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5244	376	398	518	538	527	446	431	426	421	427	377	361	(10)
(11)		CDD18.3	2202	117	164	260	288	268	196	173	167	171	169	127	102	(11)
(12)		CDH23.3	15677	659	1334	2654	3092	2547	1256	804	722	881	817	508	403	(12)
(13)		CDH26.7	5149	90	453	1210	1438	1078	323	124	83	142	132	49	25	(13)
(14)	Wind	WSAvg	1.5	1.3	1.2	1.1	1.1	1.5	2.5	2.5	2.1	1.5	1.0	1.1	1.2	(14)
(15)	Precipitation	PrecAvg	1013	2	6	13	55	120	95	110	146	215	166	57	20	(15)
(16)		PrecMax	1641	23	58	101	324	242	228	286	387	517	606	236	119	(16)
(17)		PrecMin	646	0	0	0	0	1	7	18	45	33	3	1	0	(17)
(18)		PrecStd	221	5	13	24	60	50	58	71	83	115	103	55	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	33.1	34.7	35.5	35.0	32.4	30.0	29.5	30.1	30.1	29.2	28.5	(19)
(20)			MCWB	17.2	17.4	18.4	20.4	21.0	22.5	21.2	21.4	21.1	19.5	18.4	17.2	(20)
(21)		2%	DB	28.6	31.6	33.8	34.6	33.9	30.7	29.0	28.4	29.1	29.0	28.0	27.4	(21)
(22)			MCWB	17.1	17.4	18.4	20.2	21.3	22.3	21.5	21.5	21.0	19.9	18.8	17.5	(22)
(23)		5%	DB	27.7	30.5	33.0	33.8	32.9	29.6	28.1	27.6	28.3	28.1	27.1	26.5	(23)
(24)			MCWB	16.9	17.2	18.4	20.1	21.6	22.0	21.3	21.3	21.0	20.3	19.1	17.5	(24)
(25)		10%	DB	26.8	29.5	32.1	32.8	31.7	28.6	27.1	26.8	27.4	27.3	26.2	25.7	(25)
(26)			MCWB	16.8	17.2	18.1	20.3	21.8	21.8	21.2	21.2	21.0	20.5	19.0	17.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.5	22.8	24.0	24.5	24.2	23.1	23.2	23.2	23.0	22.2	21.2	(27)
(28)			MCDWB	25.5	27.2	29.1	30.1	30.3	29.3	27.2	27.1	27.0	26.8	25.5	24.4	(28)
(29)		2%	WB	19.9	20.7	22.0	23.2	23.7	23.4	22.5	22.4	22.5	22.4	21.5	20.5	(29)
(30)			MCDWB	24.6	26.3	27.8	29.3	29.5	28.5	26.7	26.3	26.4	25.9	24.6	23.6	(30)
(31)		5%	WB	19.3	20.1	21.4	22.7	23.3	22.8	22.0	22.0	22.1	22.0	21.1	20.0	(31)
(32)			MCDWB	24.0	25.5	27.4	28.7	29.1	27.7	26.2	25.8	25.9	25.4	24.1	23.0	(32)
(33)		10%	WB	18.8	19.6	20.9	22.2	22.9	22.3	21.6	21.6	21.7	21.6	20.6	19.5	(33)
(34)			MCDWB	23.3	24.9	27.0	28.1	28.6	26.8	25.6	25.2	25.4	24.8	23.6	22.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	11.3	11.2	10.3	9.6	7.5	6.8	6.5	7.0	7.1	7.5	8.8	(35)
(36)			MCDBR	11.3	12.3	11.8	11.1	11.0	9.0	8.2	7.6	8.3	8.5	9.3	10.1	(36)
(37)			MCWBR	3.0	3.3	3.9	4.0	3.3	2.7	2.5	2.4	2.4	2.6	2.8	2.9	(37)
(38)		5% WB	MCDWB	8.8	9.7	10.0	10.1	9.7	8.0	7.1	6.7	6.9	6.6	6.3	6.9	(38)
(39)			MCWBR	3.1	3.3	4.0	3.9	3.6	3.1	2.7	2.6	2.6	2.6	2.6	2.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.383	0.404	0.466	0.508	0.522	0.454	0.466	0.454	0.440	0.472	0.413	0.389	(40)	
(41)		taud	2.306	2.251	2.084	2.013	2.003	2.254	2.210	2.250	2.284	2.169	2.313	2.332	(41)	
(42)		Ebn at Noon	914	908	857	815	789	836	828	848	867	833	878	898	(42)	
(43)		Edh at Noon	128	140	169	181	179	138	145	141	136	150	126	122	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.41	6.18	6.54	6.57	6.26	5.23	4.66	4.73	5.16	4.96	4.62	4.74	(44)	
(45)		RadStd	0.24	0.25	0.22	0.21	0.31	0.38	0.29	0.29	0.29	0.35	0.42	0.33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.36	+0.39	N/A	+0.50	N/A	N/A	N/A	N/A	+171	+172
(47) Regional (3 neighbors)		+0.32	N/A	N/A	+0.52	N/A	N/A	N/A	N/A	+171	+172

Nomenclature: See separate page